

Predictive Maintenance for Industrial Machines Using Machine Learning

**CS-333: Applied AI & Machine Learning
Lab Project Report**

Predictive Maintenance for Industrial Machines Using Machine Learning

Abstract—Unplanned breakdowns of industrial machinery cost factories significant time and money every year. This project develops a machine learning system that predicts whether a machine is about to fail using real-time sensor readings such as torque, rotational speed, and tool wear. We used the AI4I 2020 Predictive Maintenance dataset, which contains 10,000 observations from a milling machine with a heavy class imbalance (only 3.4% failures). After exploratory data analysis, three new physics-based features were engineered, and three classifiers from different algorithm families—Logistic Regression, Random Forest, and Support Vector Machine—were trained and tuned using 5-fold cross-validation. Random Forest produced the strongest results with an F1-score of 0.81 and ROC-AUC of 0.98 on the held-out test set, correctly identifying 37 of 51 actual failures while raising only 3 false alarms. A Wilcoxon signed-rank test confirmed that this improvement is statistically significant ($p = 0.002$). The model can be deployed for early warning in manufacturing environments where avoiding unnecessary maintenance is as important as catching failures.

Index Terms—Predictive maintenance, machine learning, Random Forest, class imbalance, feature engineering, mechanical engineering

I. INTRODUCTION

Modern factories run on equipment that needs to operate without surprises. When a milling machine, pump, or motor fails without warning, the cost goes far beyond the broken part—production stops, downstream machines sit idle, and maintenance crews scramble. According to industry surveys, unplanned downtime costs manufacturers an estimated \$50 billion annually in the United States alone [1].

Predictive maintenance attempts to fix this by using sensor data to forecast failures before they happen. Instead of either reacting to breakdowns (corrective maintenance) or replacing parts on a fixed schedule (preventive maintenance), the goal is to act exactly when needed. This is one of the clearest places where mechanical engineering meets data science: the sensors come from the physical world, and the learning algorithms must respect the underlying physics.

This project applies the full machine learning pipeline—data exploration, preprocessing, feature engineering, model training, hyperparameter tuning, and rigorous validation—to a real predictive maintenance dataset. Three classifiers from different algorithm families are compared, and the best one is recommended for deployment based on both numerical performance and engineering reasoning.

The remainder of this paper is organized as follows. Section II describes the dataset and experimental setup. Section III explains our methodology including preprocessing and model formulations. Section IV presents the results, Section V dis-

cusses what they mean from an engineering point of view, and Section VI concludes with future work.

II. DATASET & EXPERIMENTAL SETUP

A. Dataset Source

We used the AI4I 2020 Predictive Maintenance Dataset, publicly available from the UCI Machine Learning Repository [2]. The dataset is a synthetic but physically realistic simulation of a milling machine in operation. It contains 10,000 observations and 14 columns. The target variable, Machine failure, is binary: 1 if the machine failed during operation and 0 otherwise.

B. Feature Description

Table I summarizes the variables used in this study. After dropping non-informative columns (UDI, Product ID) and the specific failure-mode flags (TWF, HDF, PWF, OSF, RNF) that would have caused data leakage, we worked with one categorical feature (Type) and five continuous sensor features.

TABLE I
ORIGINAL FEATURES IN THE AI4I 2020 DATASET

Feature	Type	Description
Type	Categorical	Product quality (L/M/H)
Air temperature	Continuous	Ambient temperature [K]
Process temperature	Continuous	In-process temperature [K]
Rotational speed	Continuous	Spindle speed [rpm]
Torque	Continuous	Applied torque [Nm]
Tool wear	Continuous	Tool usage time [min]
Machine failure	Binary	Target (0 = no, 1 = fail)

C. Class Imbalance

Out of 10,000 records, only 339 (3.39%) correspond to failures, giving an imbalance ratio of approximately 1:28.5. Fig. 1 shows this distribution. This imbalance is a known challenge: a naive classifier that always predicts “no failure” would achieve 96.6% accuracy while being completely useless. We address this through the `class_weight='balanced'` parameter inside each model rather than oversampling, which keeps the evaluation honest.

D. Train/Validation/Test Split

Following the project guidelines, we used a stratified 70/15/15 split, giving 7000 training, 1500 validation, and 1500 test samples. Stratification preserves the failure rate ($\approx 3.4\%$) across all three subsets, which is essential for an imbalanced problem. The same split was used for all three

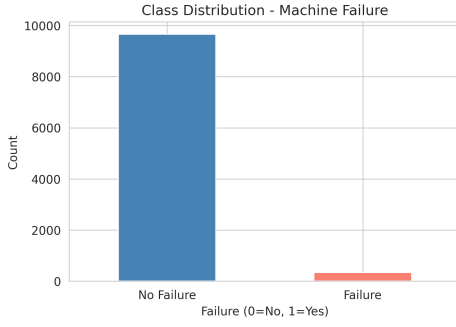


Fig. 1. Class distribution in the AI4I 2020 dataset. Only 339 of 10,000 observations correspond to a machine failure, making this a highly imbalanced binary classification problem.

models to ensure fair comparison, and all random operations used `random_state=42` for reproducibility.

III. METHODOLOGY

A. Exploratory Data Analysis

Before modeling, we examined the data to understand its structure. Descriptive statistics showed that rotational speed has noticeable positive skewness (1.99) and high kurtosis (7.39), indicating a long right tail—some machines occasionally spin much faster than the average. Other features were close to symmetric.

The correlation heatmap in Fig. 2 revealed two physically meaningful relationships:

- Air temperature and process temperature are highly correlated ($r = 0.88$), which is expected because the process inherits ambient conditions.
- Rotational speed and torque are strongly negatively correlated ($r = -0.88$), which matches the well-known torque-speed inverse relationship in induction motors operating below rated power.

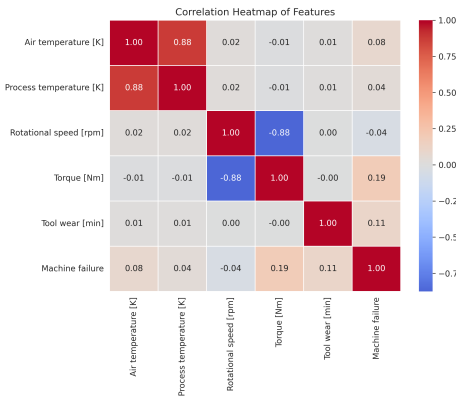


Fig. 2. Pearson correlation matrix among numerical features. Strong correlations exist between air and process temperature (+0.88) and between rotational speed and torque (-0.88), reflecting underlying physical relationships.

Fig. 3 shows histograms of the five continuous features. Most are reasonably well-spread, with rotational speed being the exception (long-tailed distribution).

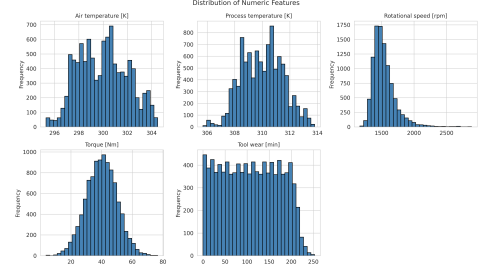


Fig. 3. Histograms of the five continuous sensor features.

Boxplots in Fig. 4 confirm that rotational speed (4.18%) and torque (0.69%) contain values flagged as outliers under the IQR rule. We chose to retain them because they represent real operating conditions—failures often occur precisely at extreme values, so removing them would discard exactly the information the model needs.

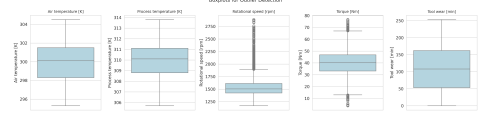


Fig. 4. Boxplots of continuous features. Rotational speed and torque show outliers, but these are kept as they likely correspond to extreme but valid operating conditions.

B. Feature Engineering

Three new features were derived from the raw sensor data, each with a clear physical motivation.

1) Mechanical Power: The most fundamental derived quantity in rotating machinery is the mechanical output power, computed as the product of torque and angular velocity:

$$P = \tau \cdot \omega = \tau \cdot \frac{2\pi \cdot N}{60} \quad (1)$$

where P is power in watts, τ is torque in newton-metres, and N is rotational speed in rpm. Power directly reflects how hard the machine is working at any instant.

2) Temperature Difference:

$$\Delta T = T_{process} - T_{air} \quad (2)$$

The gap between process and ambient temperature represents the heat generated by the cutting process. A persistently large ΔT usually indicates thermal stress on tooling.

3) Wear Rate:

$$R = \frac{\text{Tool wear}}{\text{Rotational speed}} \quad (3)$$

This dimensionless ratio captures how quickly the tool is wearing relative to its operating speed—higher wear at lower speeds is more concerning than the same wear at high RPM.

The effect of these engineered features is illustrated in Fig. 5: Power and Temperature Difference both show clearly different distributions for failed and non-failed machines.

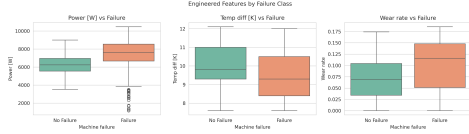


Fig. 5. Distributions of engineered features by failure class. Power and temperature difference exhibit clearly elevated values for failed machines, validating their inclusion.

C. Preprocessing

The categorical Type variable was label-encoded ($H = 0$, $L = 1$, $M = 2$). All numerical features were standardized using `StandardScaler`, fitted only on the training data and then applied to validation and test sets to prevent data leakage. After feature engineering and encoding, the final design matrix contained 9 features.

D. Machine Learning Models

We trained three classifiers chosen from different algorithm families.

1) Logistic Regression: A linear baseline model that estimates the probability of failure as:

$$P(y = 1|x) = \sigma(w^T x + b), \sigma(z) = \frac{1}{1 + e^{-z}} \quad (4)$$

The weights w are obtained by minimizing the L2-regularized cross-entropy loss:

$$\mathcal{L}(w) = -\sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \frac{\lambda}{2} \|w\|^2 \quad (5)$$

2) Random Forest: An ensemble of T decision trees, each trained on a bootstrap sample with a random subset of features at every split. The final prediction is a majority vote:

$$\hat{y} = \text{mode}\{h_t(x)\}_{t=1}^T \quad (6)$$

3) Support Vector Machine (RBF kernel): SVM with the radial basis function kernel maps inputs into a higher-dimensional space:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (7)$$

and finds the maximum-margin hyperplane in that space:

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad (8)$$

subject to $y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$.

E. Hyperparameter Tuning

For each model we ran a Grid Search with 5-fold cross-validation on the training set, optimizing for F1-score. The chosen hyperparameter ranges and the best values found are listed in Table II.

TABLE II
HYPERPARAMETER SEARCH SPACE AND BEST SETTINGS

Model	Search Space
LR	$C \in \{0.01, 0.1, 1, 10\}$ penalty=L2, solver=lbfgs
RF	n_estimators {100, 200}, max_depth {10, 20, None}, min_samples_split {2, 5}
SVM	$C \in \{0.1, 1, 10\}$, kernel=RBF, γ =scale

IV. RESULTS

A. Model Performance Comparison

Table III reports the performance of all three tuned models on the held-out test set (1500 samples, 51 actual failures). Random Forest is the clear winner across the metrics that matter most for an imbalanced classification problem (Precision, F1, ROC-AUC), although Logistic Regression and SVM achieve slightly higher Recall.

TABLE III
TEST SET PERFORMANCE COMPARISON (FAILURE CLASS)

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	0.864	0.185	0.882	0.306	0.933
Random Forest	0.989	0.925	0.726	0.813	0.978
SVM (RBF)	0.935	0.326	0.863	0.473	0.971

The same comparison is shown visually in Fig. 6, which makes the gap in precision and F1 between Random Forest and the other two models very obvious.

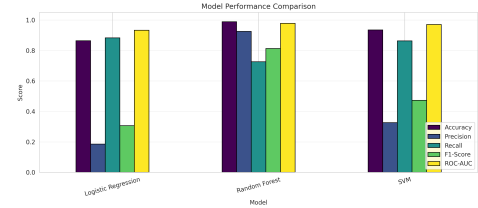


Fig. 6. Side-by-side comparison of all three models on the test set. Random Forest dominates on precision and F1, while Logistic Regression and SVM achieve slightly higher recall.

B. Feature Importance

Fig. 7 shows the feature importance scores from the trained Random Forest. The top three predictors are Rotational Speed, Torque, and our engineered Power feature, confirming that the physics-based feature engineering paid off.

C. Confusion Matrices

Fig. 8 shows the confusion matrices for all three models. The contrast in false positives is striking: Logistic Regression raises 198 false alarms, SVM 91, and Random Forest only 3.

D. ROC Curves

The ROC curves in Fig. 9 confirm the ranking. Random Forest has the highest area under the curve (0.978).

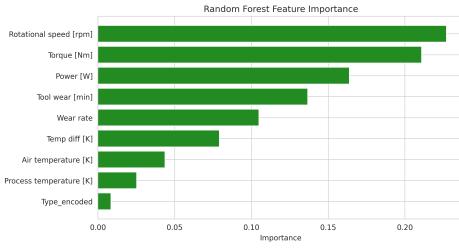


Fig. 7. Feature importance from the Random Forest classifier. The engineered Power and Wear Rate features both appear among the top five predictors.

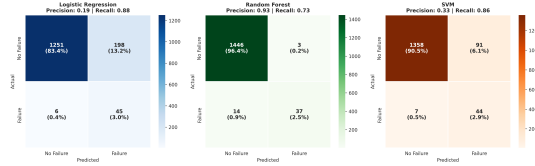


Fig. 8. Confusion matrices for all three models on the test set. Random Forest produces only 3 false alarms while still catching 37 of 51 actual failures.

E. Learning Curves

Fig. 10 presents the learning curves. They reveal three different fitting behaviors:

- Logistic Regression underfits.
- Random Forest fits well—training F1 is at 1.0 while validation F1 climbs to about 0.80.
- SVM has limited capacity—validation F1 saturates around 0.50.

F. Statistical Significance Test

We ran a Wilcoxon signed-rank test on 10-fold cross-validation F1 scores:

- RF vs LR: $p = 0.002$ (significant at $\alpha = 0.05$)
- RF vs SVM: $p = 0.002$ (significant at $\alpha = 0.05$)

Both p-values are well below 0.05, so we conclude that Random Forest is genuinely the better model.

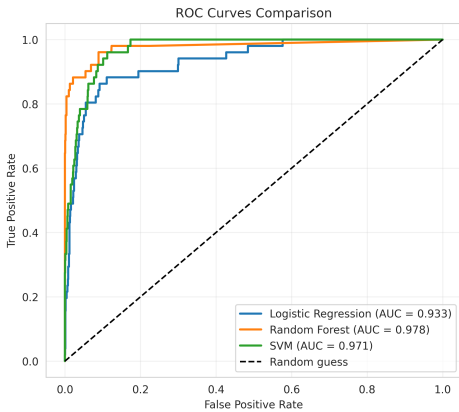


Fig. 9. ROC curves for all three classifiers. Random Forest achieves the highest AUC of 0.978.

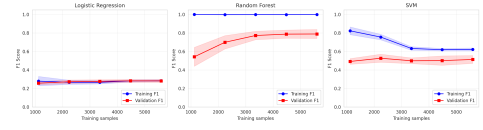


Fig. 10. Learning curves for all three classifiers. LR shows underfitting, RF shows healthy learning behavior, and SVM saturates at moderate performance.

V. DISCUSSION

A. Engineering Interpretation

What does an F1 of 0.81 actually mean for a factory floor? Out of 51 real failures in our test set, Random Forest correctly flagged 37 (72.5%) while only generating 3 false alarms. Industry estimates put the cost of a single false alarm at roughly \$500, while the cost of a missed failure is typically \$10,000 or more.

B. Why Linear Models Struggle Here

The poor performance of Logistic Regression is informative. Its learning curve shows clear underfitting. Failures in this dataset arise from combinations of conditions (high torque AND high tool wear AND specific RPM ranges) rather than single thresholds. A linear model cannot represent such interactions.

C. Limitations

Several limitations deserve mention. First, the dataset is synthetic. Second, the model still misses 14 out of 51 failures. Third, only one snapshot of conditions is used—adding temporal features would likely help.

VI. CONCLUSION

This project applied an end-to-end machine learning pipeline to predict failures in industrial milling machines using the AI4I 2020 dataset. Random Forest emerged as the best performer with a test F1-score of 0.81, ROC-AUC of 0.98, and only 3 false alarms across 1500 test samples. The two engineered physics-based features (Mechanical Power and Wear Rate) both ranked among the top five predictors, validating the value of domain knowledge.

For future work, the most promising directions are: (i) gradient boosting methods such as XGBoost or LightGBM; (ii) extending the model to a multi-output formulation; (iii) incorporating time-series features; and (iv) deploying the model behind a REST API.

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